On compression paradigms

On-disk compression VS random-access compression

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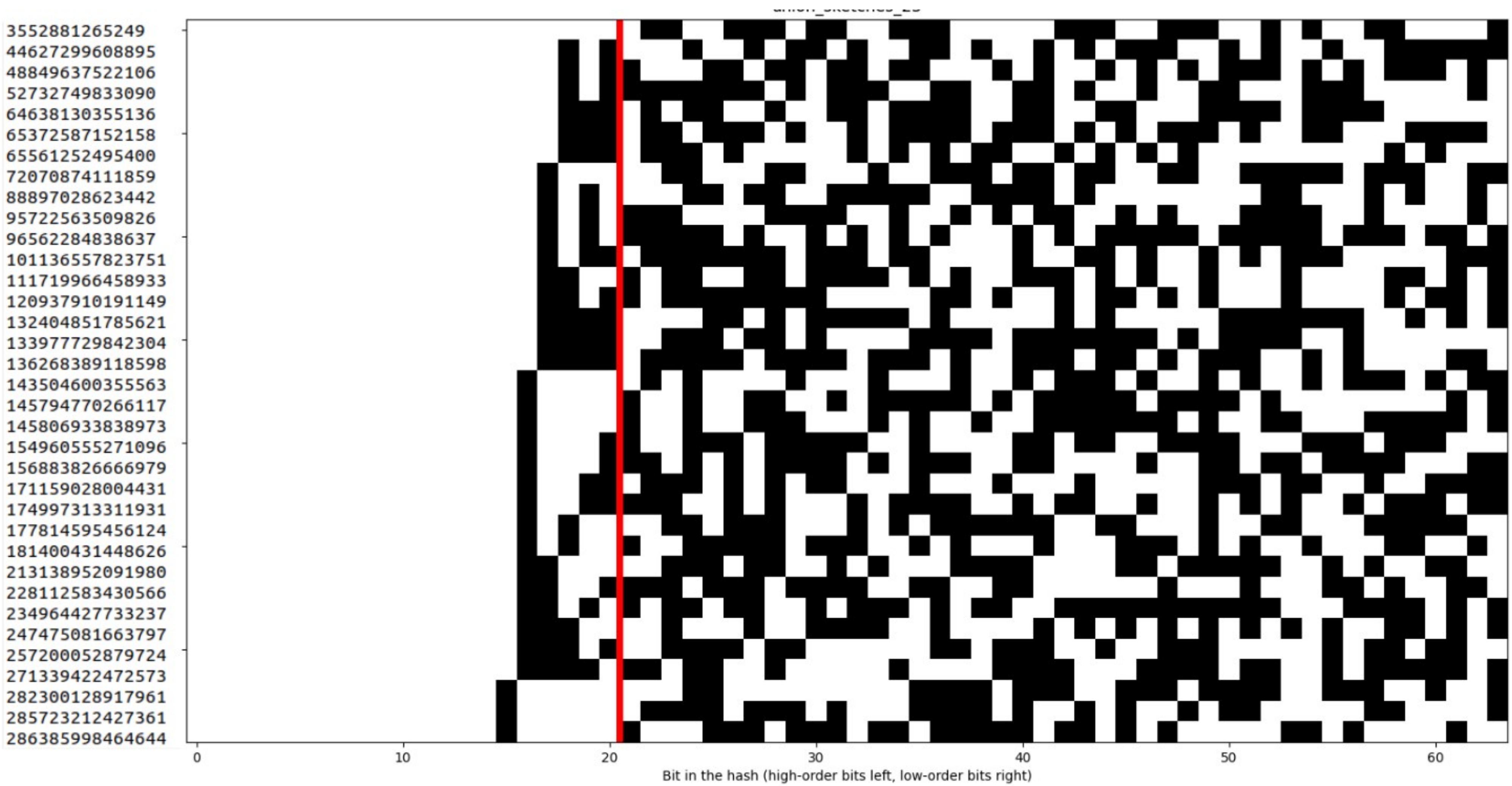
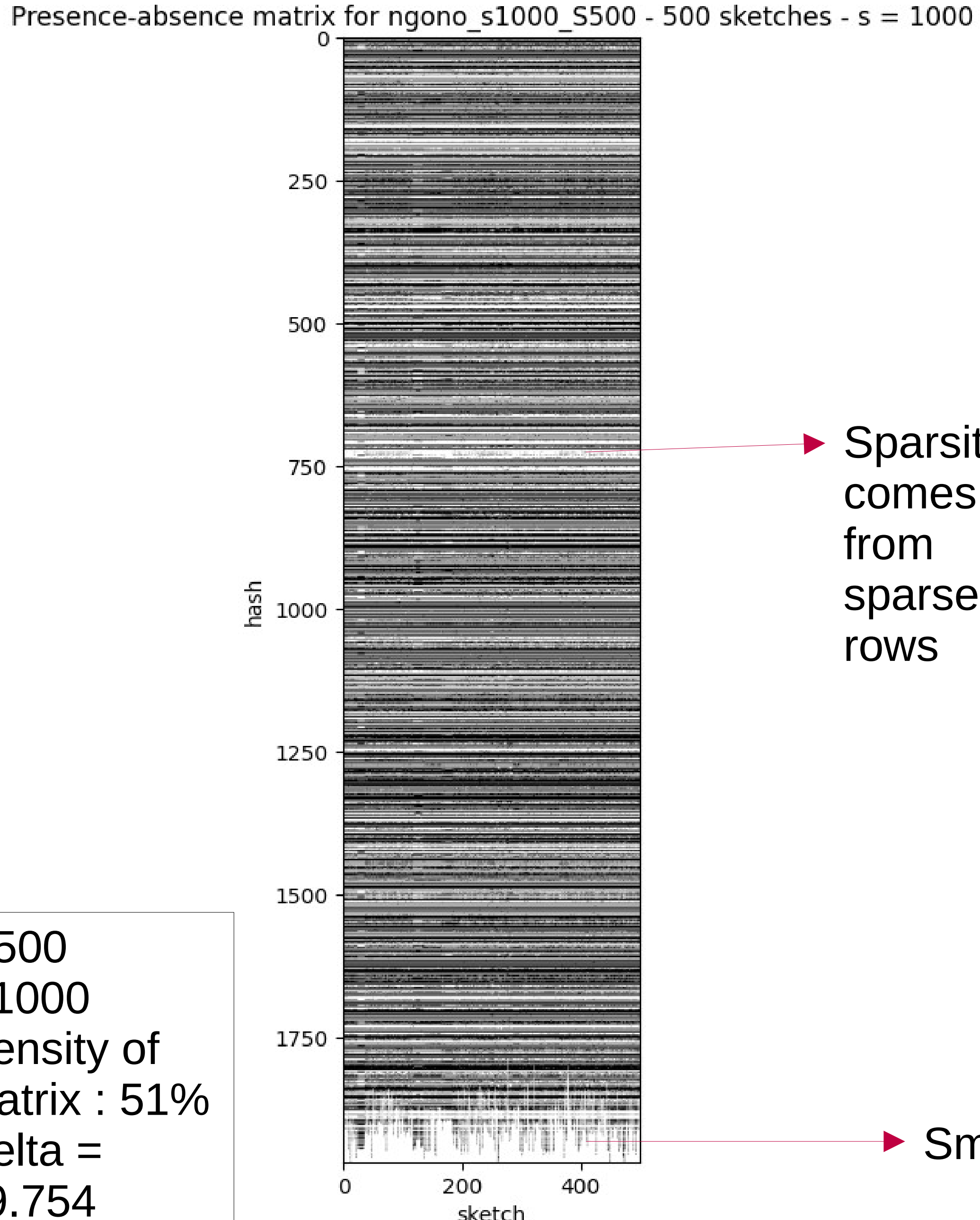
Quick recap of last time's results

Compression of the union of sketches : Elias-Fano



$$\approx 2\mathcal{N}_h + \mathcal{N}_h \log_2\left(\frac{m}{\mathcal{N}_h}\right) \text{ bits}$$

Quick recap of last time's results



Sparsity comes from sparse rows

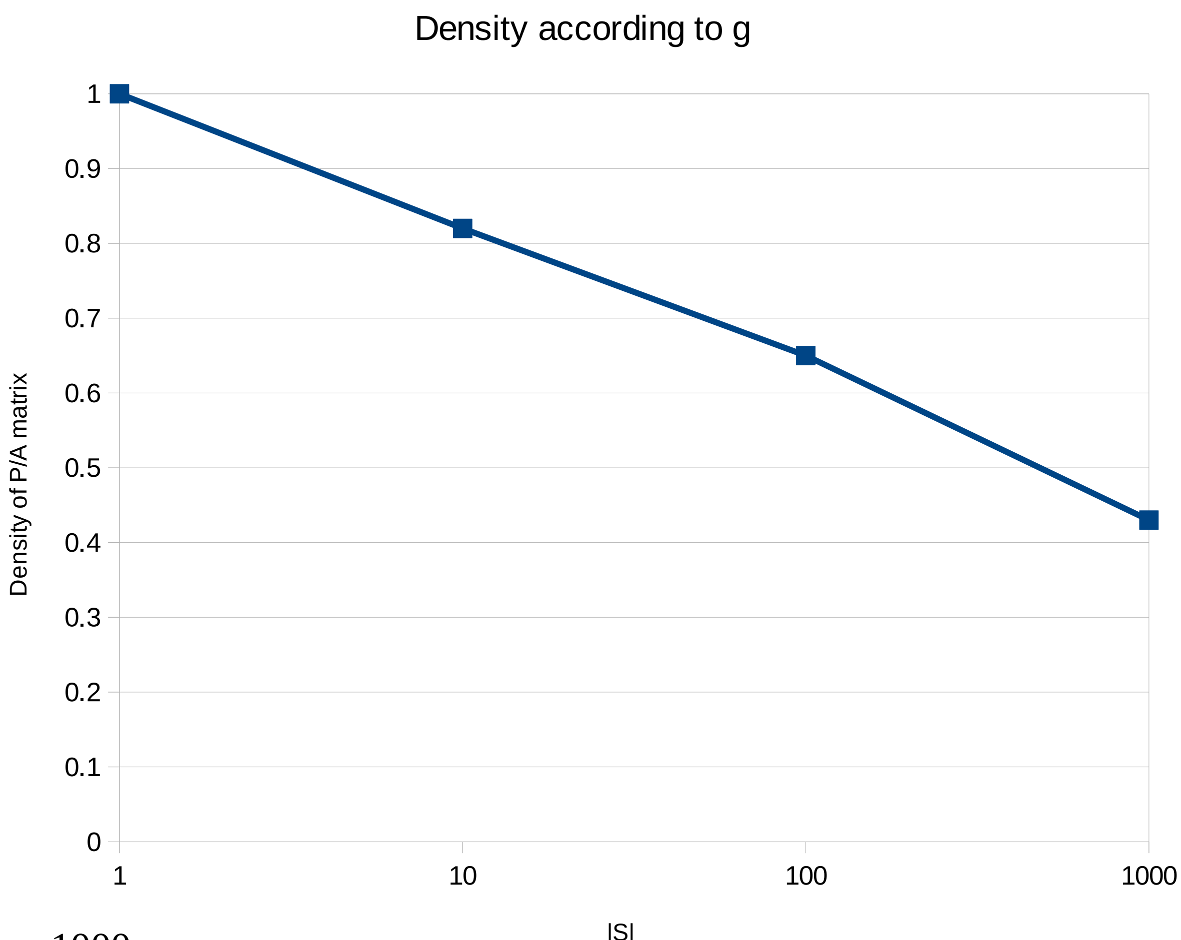
Small sparse area at the end

s	sketch size
g	number of genomes
δ	$\frac{1}{2}\sum_{i=1}^{g-1} d_{Hamming}(S(g_i), S(g_{i+1}))$
$\mathcal{N}_h$	number of distinct hashes
$\mathcal{S}_U$	union of all sketches
$\mathcal{M}$	presence matrix of hashes
d_1	density of ones in $\mathcal{M}$

g = 500
s = 1000
- Density of matrix : 51%
- Delta = 29.754

Quick recap of last time's results

Density of ones in the presence/absence matrix



s	sketch size
g	number of genomes
δ	$\frac{1}{2}\sum_{i=1}^{g-1} d_{Hamming}(S(g_i), S(g_{i+1}))$
$\mathcal{N}_h$	number of distinct hashes
$\mathcal{S}_U$	union of all sketches
$\mathcal{M}$	presence matrix of hashes
d_1	density of ones in $\mathcal{M}$

- s = 1000
- Neisseria gonorrhoeae

Density of ones

Density of ones

Let d_1 be the density of ones of the presence matrix.

$$d_1 = \frac{\text{nb ones}}{\text{matrix size}} = \frac{sg}{g\mathcal{N}_h} = \frac{s}{\mathcal{N}_h}$$

s	sketch size
g	number of genomes
δ	$\frac{1}{2}\sum_{i=1}^{g-1} d_{Hamming}(S(g_i), S(g_{i+1}))$
$\mathcal{N}_h$	number of distinct hashes
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Trivially

$$s + \delta \geqslant \mathcal{N}_h \geqslant s$$

Upper boud rached by having distinct hashes values for all changes.
Get a finer lower bound :

$$\mathcal{N}_h \geqslant \frac{2\delta}{g-1}$$

Conjecture : $\mathcal{N}_h \geqslant s + \frac{\delta}{g}$

$$\min\left(1, \frac{sg}{2\delta}\right) \geqslant d_1 \geqslant \frac{s}{s+\delta} = \frac{1}{1+\frac{\delta}{s}}$$

$$\frac{2}{g} r \geqslant \frac{1}{1+\frac{\delta}{s}}$$

Compression paradigms

On-disk compression :

- Immutable data structure
- Input : list of sketches of size s
- Query : get the list of sketches

Random-access compression

- (Im?)mutable data structure
- Input : list of sketches of size s
- Query : get the i^{th} sketch in the list

Basic compression algorithm

```
Input : list [S_1, … S_g] of
sketches of size s
```

- `U ← sorted union of S_1,…,S_g`
- `M ← presence matrix of hashes in sketches`
- `Compress U with Elias-Fano`
- `Compress M with RLE`
- `Return (U, M)`

Does not have to be explicit (to save RAM : s = 10000, g = 4000, ngono : |M| : 18Mb)

Or another compression tool

Random access : query

- Retrieve the ith column of M
- Deduce its hashes

Linear in M, if RLE used :
Auxiliary structure that stores positions ?

S = 10,000
G = 4000
Ngono
→ |M| = 18Mb
Other compressed representation

Conclusions

- Look into another compression scheme for M for random access and on-disk compression
- Binary matrix structure close to a full rectangle with sparse rows means compressibility might be expressed using the density